EXP NO - 5: SCHEDULING ALGORITHMS (ROUND ROBIN SCHEDULING)

```
[rene@localhost ~]$ ls
abc.sh     Desktop    filename.txt  newfile.c   producerconsumer  text
a.out      Documents  file.sh       page        Public            text.c
bank       Downloads  firstfit.c    p.c         rec               Videos
bankers    e.sh       first.sh      Pictures    scheduling        xyz
bankers.c  exec.c     memory        pid.c       sjf.c
cd         f.c        Music         pro         system_calls
ch         fcfs.c     name.sh       producer.c  Templates
[rene@localhost ~]$ cd scheduling
[rene@localhost scheduling]$ ls
a.out  priority.c
[rene@localhost scheduling]$ vi rr.c
[rene@localhost scheduling]$
```

```c
#include<stdio.h>
int main(){
        int n,tq,completed=0,completionTime =0;
        printf("Enter number of processes: \n");
        scanf("%d", &n);
        printf("Enter Time Quantum: \n");
        scanf("%d" ,&tq);
        int at[n],bt[n],rem[n],tat[n],wt[n];
        float avgtat = 0,avgwt = 0;
        for(int i = 0; i < n; i++){
                printf("Enter AT and BT for P%d: \n", i + 1);
                scanf("%d %d" ,&at[i] ,&bt[i]);
                rem[i] = bt[i];
        }
        while(completed < n)
        {
                int idle = 1;
                for(int i=0;i<n;i++)
                {
                        if(at[i] <= completionTime && rem[i] > 0)
                        {
                                idle = 0;
                                if(rem[i] > tq)
                                {
                                        completionTime += tq;
                                        rem[i] -= tq;
                                }else{
                                        completionTime += rem[i];
                                        tat[i] = completionTime - at[i];
                                        wt[i] = tat[i] - bt[i];
                                        rem[i] = 0;
                                        completed++;
                                }
                        }

                }
                if(idle){
                        completionTime++;
                }
        }
        for(int i=0;i<n;i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }
                                                                                    1,17          Top
```

```c
        scanf("%d", &n);
        printf("Enter Time Quantum: \n");
        scanf("%d" ,&tq);
        int at[n],bt[n],rem[n],tat[n],wt[n];
        float avgtat = 0,avgwt = 0;
        for(int i = 0; i < n; i++){
                printf("Enter AT and BT for P%d: \n", i + 1);
                scanf("%d %d" ,&at[i] ,&bt[i]);
                rem[i] = bt[i];
        }
        while(completed < n)
        {
                int idle = 1;
                for(int i=0;i<n;i++)
                {
                        if(at[i] <= completionTime && rem[i] > 0)
                        {
                                idle = 0;
                                if(rem[i] > tq)
                                {
                                        completionTime += tq;
                                        rem[i] -= tq;
                                }else{
                                        completionTime += rem[i];
                                        tat[i] = completionTime - at[i];
                                        wt[i] = tat[i] - bt[i];
                                        rem[i] = 0;
                                        completed++;
                                }
                        }

                }
                if(idle){
                        completionTime++;
                }
        }
        for(int i=0;i<n;i++)
        {
                avgtat += tat[i];
                avgwt += wt[i];
        }
        printf("\nAverage Turnaround Time: %.2f\n", avgtat / n);
        printf("Average Waiting Time: %.2f\n", avgwt / n);
        return 0;
}
                                                                                                                    49,1          Bot
```

```
[rene@localhost ~]$ ls
abc.sh     Desktop    filename.txt  newfile.c   producerconsumer  text
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bank       Downloads  firstfit.c    p.c         rec               Videos
bankers    e.sh       first.sh      Pictures    scheduling        xyz
bankers.c  exec.c     memory        pid.c       sjf.c
cd         f.c        Music         pro         system_calls
ch         fcfs.c     name.sh       producer.c  Templates
[rene@localhost ~]$ cd scheduling
[rene@localhost scheduling]$ ls
a.out  priority.c  rr.c
[rene@localhost scheduling]$ vi rr.c
[rene@localhost scheduling]$ gcc rr.c
[rene@localhost scheduling]$ ./a.out
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 4
Enter AT and BT for P3:
2 2
Enter AT and BT for P4:
4 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
[rene@localhost scheduling]$
```